

Nicholas Junge

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Mathematician and software engineer with 5+ years of experience in ML, building open-source tooling and on-device inference pipelines. Independent work in scientific computing, GPU kernels, and compilers.

Work Experience

Senior MLOps Engineer, appliedAI Institute for Europe gGmbH (Munich, DE) Nov 2022 – present

Developed [lakefs-spec](#) for data versioning and [nnbench](#) for model evaluation (>4k installs/month across both).

Currently building on-device pipelines for landscape detail detection in aerial imagery with SAM v1/2, MLX, and BentoML; created a QGIS plugin for the project. Part of the team placing second in the [KI Kommune 2024](#) hackathon.

Tools: Python, C++, SAM v1/2, QGIS, Docker, Kubernetes, MLflow, JAX, MLX, BentoML.

Data Engineer / Analyst, Junge Die Bäckerei GmbH (Lübeck, DE-remote) Jul 2021 – Apr 2026

Advised on a demand forecasting project, and maintained ETL pipelines for review aggregation and data quality assurance.

Tools: Python, Windows Server, MSSQL, Prefect, scikit-learn, duckdb.

Working Student ML Engineer, ZenML GmbH (Munich, DE) Jun 2020 – Mar 2021

Built a test suite with >80% coverage and ML pipeline components for ZenML's launch; ran distributed training for predictive maintenance on terabytes of bus data on GCP.

Tools: Python, Apache Beam, TensorFlow, TFX, BigQuery, Google Cloud Vertex AI.

Working Student Modeling & Simulation, Siemens (Munich, DE) Aug 2017 – Sep 2019

Implemented a C++17 ODE solver for crack propagation and integrated it with Siemens NX CAD; built a real-time ML app with TensorFlow to predict transformer current loss.

Tools: Python, C++17, React, TensorFlow, Pandas, Websockets, Siemens NX.

Selected Projects and Contributions

[Palladium](#) and [metal-runtime](#) — Compilers and GPU runtimes Independent projects

Built a compiler that translates JAX/Pallas GPU kernels to Metal, and a C++ runtime with native JAX FFI dispatch, validation against CPU reference results, and batched/concurrent execution.

On an Apple M1 Pro, this runs an ensemble of 100,000 independent simulations in JAX 40x faster than the CPU-only baseline.

- **Benchmarking:** Maintaining Google Benchmark's Python bindings; received a Google Open Source Peer Bonus in 2023. Built [mew](#) on top of it with a Python API, CLI, and comparison workflow.
- **Open source:** Contributed NumPy functionality and numerical solver improvements to [JAX](#). Also contributed to MLIR, DuckDB, and nanobind. Authored [nanobind-bazel](#) for creating C++-to-Python bindings in Bazel.
- **Speaking:** Presented a RAG-based news aggregation Slackbot project with Langfuse observability at the MLOps Community Meetup, Munich (October 2025).

Education

M.Sc. Mathematics | TUM (Munich, DE) | GPA (US/DE): 3.5/1.5 Oct 2017 – Jun 2020

Study topics: Mathematical modeling, statistics and machine learning, time series analysis, optimization.

B.Sc. Physics | Bielefeld University (Bielefeld, DE) | GPA (US/DE): 3.2/1.8 Oct 2013 – Sep 2017

Study topics: Theoretical physics, numerical analysis, ODEs/PDEs. Semester abroad at **Bergen University**.

Skills, Tools & Languages

- **Programming:** Python, C++17, TypeScript, Metal.
- **Frameworks & tools:** JAX/Pallas, MLX, BentoML, nanobind, CMake, Bazel, Docker, Kubernetes, Terraform.
- **Spoken languages:** German, English, Norwegian, Danish, Spanish.